



MSc Group Design Handbook

CADEM0016

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Revision History

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Executive Summary

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1 Introduction

In this project, students will work in teams to develop a multidisciplinary design optimisation (MDO) tool for the conceptual assessment of multiple aircraft configurations.

The project is organised into four iterative sprints, using the agile concept of a sprint to structure each stage of work:

- **Down-Selection:** Teams use engineering judgement to compare candidate aircraft concepts and select two planforms for further study. One of these must be a conventional tube-and-wing configuration capable of replicating the baseline aircraft (the Boeing 777F), which serves as the project reference.
- **Development:** Team members develop discipline-specific class I/II tools to develop an MDO framework which is capable of sizing both of the configurations chosen in the previous sprint. At the conclusion of this sprint, the team must evaluate results and select a single concept to carry forward.
- **Refinement:** The team advances discipline-specific tools to class II.5 fidelity, introducing physics-based estimations calibrated against existing aircraft data.
- **Dissemination:** The team completes a Final design review presentation, submits a final report, and individual team members submit a poster and complete a short individual presentation.

A note on method classification: The classifications of methods described in the three sprints are based on Torenbeek's classification of aircraft sizing methods [4]:

- **Class I:** Statistical/empirical correlations for rapid conceptual estimates.
- **Class II:** Semi-empirical methods combining correlations with simplified physics.
- **Class II.5:** Physics-based simplified models calibrated to empirical data, bridging preliminary and detailed design.
- **Class III:** Full physics-based methods (e.g. finite element analysis) used in preliminary design.

During the project, students will begin with Class I methods provided in the baseline MDO tool, during the "Development sprint" these methods will be (in general) extended to class-II methods. Then, in the "Refinement Sprint", the methods will be further refined to Class II.5 fidelity.

Note that the Class I/II/II.5 terminology is used here as a pedagogical framework to illustrate progressive levels of model fidelity. Assessment will focus on the quality, clarity, and justification of the work produced, rather than the nominal classification level itself.

1.1 Milestones

The project is structured around four *sprints*, each with clear milestones to guide progress and ensure alignment across disciplines. This is a twenty-credit module, and as such, students

are expected to spend approximately 200 hours on this project. Over the 12-week teaching block, this equates to 16.5 hours, or 2 days, of work a week.

Table 2: Module Schedule: Assessments and Phases

Week	W.C.	Assessment	Phase
13	19/01/2026		Concept Down Selection
14	26/01/2026		
15	02/02/2026		MDO Development
16	09/02/2026		
17	16/02/2026	Conceptual Design Review (Fri)	
18	23/02/2026	(Reading Week)	
19	02/03/2026		MDO Refinement
20	09/03/2026		
21	16/03/2026		
	23/03/2026	(Easter)	
	30/03/2026	(Easter)	
	06/04/2026	(Easter)	
22	13/04/2026	FDR (Fri)	Dissemination
23	20/04/2026	FEDR & Poster Submission (Thur)	
24	27/04/2026	Poster Presentation (Wed)	

Sprint 1 – Down-Selection By the end of Sprint 1, teams will achieve the following:

- Establish a comparative scoring framework for concept down-selection.
- Select two candidate concepts for further study.
 - One of these must be a conventional tube-and-wing configuration
- Identify a set of hyper-parameters with suitable ranges.
 - Suggested hyper-parameters include: Aspect Ratio, aircraft range, cabin length, cabin diameter, cruise Mach number, and cruise altitude.

Sprint 2 – Development By the end of Sprint 2, the team will:

- Select a single final concept from the two candidates.
- Update the hyper-parameter list and refine ranges based on Class II analyses.
- Consolidate the modelling framework across disciplines.

Sprint 3 – Refinement By the end of Sprint 3, the team will:

- Advance the chosen concept to Class II.5 fidelity.

- Finalise the MDO framework and supporting discipline-specific models.

Sprint 4 – Dissemination By the end of Sprint 4, the team will have:

- Conducted intra- and inter-disciplinary trade-studies around identified "optima"
- Completed the Final Design Review (FDR) group Presentation
- Submitted the final Engineering Design report (FEDR)
- Submitted a poster and conducted a short presentation with the poster

2 Team Roles

Each student will be assigned a specialism from the list shown in Table 3, ensuring that all major technical disciplines are represented within the team. If a team has fewer members than roles, only the *Fuselage Structures Engineer* (Role 3) or the *Landing Gear Engineer* (Role 8) may be omitted. In such cases, the corresponding discipline can be covered using Class I or simplified Class II methods (as supplied in the original MDO tool).

The following subsections detail the responsibilities and deliverables associated with each role.

Table 3: Specialisms Outline for Group Design Project

#	Role Title	Primary Responsibilities
1	Project Manager / Economics Engineer	Overall coordination, Economic analysis, constraint analysis, integration of team outputs
2	Wing Structures Engineer	Wing box design, load analysis, structural layout, fuel tank integration
3*	Fuselage Structures Engineer	Fuselage and empennage structure, pressurisation loads, structural integration
4	Wing Aerodynamics Engineer	Wing planform, aerofoil selection, high-lift devices, drag polars
5	Empennage & Stability Engineer	Tail sizing, CG range, static/manoeuvre margins, trim drag
6	Performance, Propulsion & Climate	Engine selection, mission analysis, emission analysis, fuel burn estimation
7	Optimisation & Integration Engineer	System-level optimisation, trade studies, design freeze coordination, sensitivity analysis
8*	Landing Gear Engineer	Gear layout, footprint, shock absorber design, stowage, and integration

2.1 Project Manager & Economics Engineer

The Project Manager / Economics Engineer is responsible for the overall team coordination, file management, version control, and the development of the direct operating cost (DOC) model used within the MDO framework. This person will set the agenda and lead team meetings.

This person will prepare a plan that includes timelines and a clear understanding of critical paths. This person will assess and document risks in meeting deadlines and meeting specifications. They will manage the evolving design and its features, alongside coordination and control of data and information. This role must work closely with all team members to ensure integration of all sub-elements to meet the specification. Early identification of any challenges, conflicts, or inability to meet the specified criteria or performance will be key. This role will include ensuring that adequate personnel resources can manage and complete tasks and manage the scope.

2.1.1 Responsibilities

Concept Down Selection:

- Extract critical mission parameters from requirements.
- Study a unique configuration and rank according to chosen parameters.
- Engage with specialism material and explore the supplied MDO tool.

General Roles:

- Lead team meetings, post agendas, and make sure meeting minutes are recorded.
- Responsible for the overall plan, timeline, and resource allocations, critical path analysis, and risk management
- Selecting appropriate documentation tools and a document control system for files and a version control system for code
- Responsible for ensuring documents and presentations are delivered on-time and with a cohesive voice and style (including diagrams)
- Encourage to adopt an "Agile Project Management framework", such as scrum.
- Maintain team timesheets.

MDO Development: For the chosen hyperparameters and configurations:

- Develop a direct operating cost (DOC) discipline, or collaborate with the *PPC Engineer* to incorporate your DOC model into the mission analysis tool.
- Ensure the aircraft is being designed to specifications
- In collaboration with *Optimisation Engineer* develop a relationship between fuselage length, diameter, and maximum P6P pallets.
- In collaboration with the *Mass and Stability Engineer*, ensure the MDO framework can accurately represent the reference aircraft.

- In collaboration with the *Optimisation Engineer*, develop a version control system to ensure discipline contributions can be easily integrated back into the MDO framework

MDO Refinement:

- Ensure the aircraft is being designed to specifications
- In collaboration with the *Mass and Stability Engineer*, ensure the MDO framework can accurately represent the reference aircraft.
- In collaboration with the *Optimisation Engineer*, coordinate with all disciplines an MDO framework development freeze, to ensure adequate time to finalise results.

2.1.2 Deliverables

Final Engineering Design Report (FEDR)

- Responsibility for cohesion of entire document
- Description of management methodology
- Description of Document Control System
- Description of Version Control System
- Description of DOC discipline model.

Poster Presentation

- Methodology of the Direct Operating Cost (DOC) discipline developed.
- Sensitivity study of DOC to key hyperparameters (e.g., range, payload, cruise Mach).
- Cross-discipline trade study showing DOC interactions with other metrics (e.g., fuel burn, structural mass).
- Outline a major management challenge encountered (e.g., integration across disciplines) and how it was addressed.

2.1.3 Getting Started

Set up document control and version control early, using a shared repository for files and Git for code. Begin coding the Direct Operating Cost (DOC) discipline and integrate it into the MDO framework. At the same time, review Agile project management principles (e.g., Scrum) to structure team workflows and meetings effectively.

2.2 Wing Structures Engineer

The wing structures engineer is responsible for the overall sizing of the wing structure.

2.2.1 Responsibilities

Concept Down Selection:

- Extract critical mission parameters from requirements.
- Study a unique configuration and rank according to chosen parameters.
- Engage with specialism material and explore the supplied MDO tool.

MDO Development: For the chosen hyperparameters and configurations:

- Develop a class I or class II tool to estimate wing mass for both configurations.
 - develop methodology for the alternative configuration.
 - Explore alternative class I and class II tools to estimate the wing mass of the tube-and-wing design.
 - Tune methods from existing data.
 - Understand the limitations of these methods within the chosen range of hyperparameters.
- Begin to develop a Class II.5 model (a physics-based model) for wing mass estimation.
 - Calculate shear forces, bending moments, and torques for critical 'steady' load cases.
 - Estimate wingbox and secondary structure mass.

MDO Refinement:

- Refine the Class II.5 model (physics-based model) for wing mass estimation.
 - Determine rib and stiffener layout (buckling considerations).
 - Consider gust loading load cases.
 - Incorporate a 'landing case' into the peak load estimation. If available, collaborate with the *Landing gear Engineer* to define loads.
- Collaborate with the *Aerodynamics Engineer* to incorporate mass estimates for the control surfaces and high lift devices.
- Define wing fuel tank size and integration.

2.2.2 Deliverables

Final Engineering Design Report (FEDR)

- Documentation of class I/II methods used to compare the two configurations
- Documentation of class II.5 method used estimate final configuration.
- Provide sensitivity studies around the final Design Point

- Provide EI, GJ distributions of the wing at the final design point.

Poster Presentation

- Methodology of The class I/II & II.5 methods used to estimate wing mass for the final configuration
- Comparison of the I/II and II.5 estimations across a range of hyperparameters.
- Student Selected Cross-discipline sensitivity/trade study.

2.2.3 Getting Started

Please consult classical "aircraft design" textbooks ([1, 3, 4]) for class I / II methods for estimating structural mass for a given set of hyperparameters.

2.3 Fuselage Structures Engineer

The fuselage structures engineer is responsible for the overall sizing of the fuselage and empennage structure.

2.3.1 Responsibilities

Concept Down Selection:

- Extract critical mission parameters from requirements.
- Study a unique configuration and rank according to chosen parameters.
- Engage with specialism material and explore the supplied MDO tool.

MDO Development: For the chosen hyperparameters and configurations:

- Develop a class I or class II tool to estimate fuselage and empennage mass for both configurations.
 - develop methodology for the alternative configuration.
 - Explore alternative class I and class II tools to estimate the wing mass of the tube-and-wing design.
 - Tune methods from existing data.
 - Understand the limitations of these methods within the chosen range of hyperparameters.
- Begin to develop a Class II.5 model (a physics-based model) for fuselage mass estimation.
 - Calculate shear forces, bending moments, and torques for critical 'steady' load cases.
 - Estimate fuselage structural mass programmatically from these load cases.

MDO Refinement:

- Refine the Class II.5 model (physics-based model) for wing mass estimation.
 - Determine rib and stiffener layout (buckling considerations).
 - Consider gust loading load cases.
 - Consider landing loads (3m/s landing case)
- Refine Empennage mass estimation using class I/II methods.
- Define fuselage fuel tank size and integration.

2.3.2 Deliverables

Final Engineering Design Report (FEDR)

- Documentation of class I/II methods used to compare the two configurations
- Documentation of class II.5 method used estimate final configuration.

- Provide sensitivity studies around the final Design Point
- Provide EI, GJ distributions of the fuselage at the final design point.

Poster Presentation

- Methodology of The class I/II & II.5 methods used to estimate fuselage mass for the final configuration
- Comparison of the I/II and II.5 estimations across a range of hyperparameters.
- Student Selected Cross-discipline sensitivity/trade study.

2.3.3 Getting Started

Please consult classical "aircraft design" textbooks ([1, 3, 4]) for class I / II methods for estimating structural mass for a given set of hyperparameters.

2.4 Aerodynamics Engineer

The aerodynamics engineer is responsible for wing planform sizing and drag-polar estimation.

2.4.1 Responsibilities

Concept Down Selection:

- Extract critical mission parameters from requirements.
- Study a unique configuration and rank according to chosen parameters.
- Engage with specialism material and explore the supplied MDO tool.

MDO Development: For the chosen hyperparameters and configurations:

- In collaboration with the *PPC Engineer*, complete the constraint analysis discipline to size the thrust-to-weight ratio and wing loading for each configuration
 - use class I/II methods to estimate maximum C_L during landing and take-off.
- Define the main wing planform
- In collaboration with the *Empennage & Stability Engineer* define the location of the lifting surfaces.
- Develop a drag-polar estimate for each configuration during the cruise condition.
- Define baseline spanwise loadings and aerodynamic centres for the lifting surfaces.

MDO Refinement:

- Develop class II/II.5 methods to estimate drag-polars at different flight phases (landing/take-off/cruise) as a function of altitude and Mach number
 - Use the flat-plate analogy plus component build-up method to estimate CD_0 as a function of the hyperparameters.
 - collaborate with the *Wing Structures Engineer* to define the spanwise loading which maximises the Lift-to-drag ratio during cruise
 - consider tailplane trim-drag during cruise
- For the required maximum C_L during landing, define the size and complexity of the high-lift system.

2.4.2 Deliverables

Final Engineering Design Report (FEDR)

- Documentation of constraint Analysis (in collaboration with Aerodynamics Engineer)
- Documentation of class I/II methods used to compare the two configurations
- Documentation of class II.5 method used estimate final configuration.
- Provide sensitivity studies around the final Design Point

- Provide example drag-polars during cruise, take-off, and landing for the final design point.

Poster Presentation

- Methodology of your final sizing tool
- Sensitivity study of critical parameters to hyperparameters
- Cross-discipline sensitivity study.

2.4.3 Getting Started

For constraint analysis, follow the process outlined in classical "Aircraft Design" Textbooks ([1, 3, 4]). Those same textbooks are good references for planform selection and drag polar estimation.

2.5 Mass & Stability Engineer

The mass & stability engineer is responsible for the overall sizing of the empennage and control surfaces, as well as ensuring total structural masses are representative of the appropriate class of aircraft.

2.5.1 Responsibilities

Concept Down Selection:

- Extract critical mission parameters from requirements.
- Study a unique configuration and rank according to chosen parameters.
- Engage with specialism material and explore the supplied MDO tool.

MDO Development: For the chosen hyperparameters and configurations:

- Track the breakdown of the operational empty mass (OEM) and ensure the breakdown is representative of the desired class of aircraft.
- In collaboration with the *Economics Engineer*, ensure the MDO framework can accurately represent the reference aircraft.
- estimate the centre-of-mass (COM) envelope of each configuration
- In collaboration with the *Aerodynamics Engineer*, define the location of the lifting surfaces.
- Size the empennage of both configurations with appropriate tail-volume coefficients.

MDO Refinement:

- Use class II.5 tools to size the horizontal and vertical tail-plane and primary control surfaces (or equivalent surfaces on unconventional concepts)
 - Consider controllability, longitudinal stability, and required CoM range when sizing the horizontal tailplane.
 - Consider one engine inoperative (OEI) take-off, and crosswind Landings/take-off when sizing the vertical tailplane and rudder.
- Collaborate with *Structures Engineers* to ensure appropriate mass penalties are applied from control surface sizing.
- Collaborate with the *Aerodynamics Engineer* to see the effect of horizontal tailplane sizing on trim drag.

2.5.2 Deliverables

Final Engineering Design Report (FEDR)

- Documentation of class I/II methods used to compare the two configurations
- Documentation of class II.5 method used estimate final configuration.

- Provide sensitivity studies around the final Design Point
- Provide C.G. range at the final design point.
- Discuss the impact on stability and control of any aspects of the aircraft layout that are novel/non-conventional.

Poster Presentation

- Methodology of The class I/II & II.5 methods used to estimate empennage size for the final configuration
- Comparison of the I/II and II.5 estimations across a range of hyperparameters.
- Student Selected Cross-discipline sensitivity/trade study.

2.5.3 Getting Started

Textbooks on aircraft design contain information on weight, loading, and balance. The latter, along with Flight Dynamics Control notes and text books give guidance on basic stability and control and handling qualities, and provide the fundamentals that can be extended for unconventional configurations.

2.6 Performance, Propulsion & Climate Engineer

The performance, propulsion, & climate engineer is responsible for developing the mission analysis tool, which can estimate the fuel burn and climate impact of a given mission. They are also responsible for selecting power plants.

2.6.1 Responsibilities

Concept Down Selection:

- Extract critical mission parameters from requirements.
- Study a unique configuration and rank according to chosen parameters.
- Engage with specialism material and explore the supplied MDO tool.

MDO Development: For the chosen hyperparameters and configurations:

- In collaboration with the *Aerodynamics Engineer*, complete the constraint analysis discipline to size the thrust-to-weight ratio and wing loading for each configuration
- Develop mission analysis discipline to include full mission profile (including climb, descent, and approach)
- Develop a simplified climate model to estimate the climate impact of each mission, initially only considering carbon-dioxide emission.
- Calibrate/develop the supplied, simplified, propulsion model for chosen power plants, to scale mass, size, and performance as a function of required thrust.

MDO Refinement:

- Refine the climate model to estimate the climate impact for other major sources of anthropogenic forcing.
- Consider full life-cycle emissions of kerosene and sustainable aviation fuels (SAFs) in your analysis.
- Coordinate with the *Aerodynamics Engineer* to ensure appropriate drag polars are used during each stage of a mission

2.6.2 Deliverables

Final Engineering Design Report (FEDR)

- Documentation of constraint Analysis (in collaboration with Aerodynamics Engineer)
- Documentation of Mission Analysis Methodology
- Documentation of the climate impact model used to compare the two configurations
- Provide climate impact sensitivity studies around the final Design Point
- Provide the payload-range diagram for the final design point, and analyse the climate impact across the possible combinations of payload range.

Poster Presentation

- Methodology of your final sizing tool
- Sensitivity study of critical parameters to hyperparameters
- Cross-discipline sensitivity study.

2.6.3 Getting Started

For constraint analysis, follow the process outlined in classical "Aircraft Design" Textbooks ([1, 3, 4]). For mission analysis, read about the method of "weight fractions" described in by Gudmundsson [1]. For climate analysis, please refer to the discipline-specific lecture notes. A similar approach can also be seen in a paper by [2].

2.7 Optimisation & Integration Engineer

The optimisation & integration engineer is responsible for the systems-level optimisation of the design task. Using the MDO framework as a black-box, they are responsible for ensuring

2.7.1 Responsibilities

Concept Down Selection:

- Extract critical mission parameters from requirements.
- Study a unique configuration and rank according to chosen parameters.
- Engage with specialism material and explore the supplied MDO tool.
- Choose a set of Hyperparameters (such as payload mass, Mach Number, range)

MDO Development: For the chosen hyperparameters and configurations:

- In collaboration with *Project Manager* develop a relationship between fuselage length, diameter, and maximum P6P pallets.
- Using "block fuel" as the primary metric, conduct initial optimisation with the baseline MDO tool to identify trade-offs between fleet size and payload capacity.
- Refine the required range and minimum cruise Mach number to ensure all mission objectives can be achieved.
- Collaborate with all other disciplines and conduct sensitivity studies on primary metrics from each discipline, to understand the relative significance of each parameter. This analysis may inform which parameters (for example, wing mass, CD0, or engine location) require the highest fidelity of analysis.
- In collaboration with the *Project Manager*, develop a version control system to ensure discipline contributions can be easily integrated back into the MDO framework
 - Develop a set of unit tests to ensure team contributions do not "break" the core functionality of the code.

MDO Refinement:

- Using direct-operating-cost (DOC), average temperature response (ATP), and block fuel as the primary metrics, explore a family of "optimal" designs.
- Around Single objective optima, conduct sensitivity analysis from primary metrics from each discipline to:
 - qualify the robustness of the optima in the face of likely uncertainties
 - Understand key inter-disciplinary trade-offs
- track the iteration of "optimal" designs as the MDO framework is updated.
- In collaboration with the *Project Manager*, coordinate with all disciplines an MDO development freeze, to ensure adequate time to finalise results.

2.7.2 Deliverables

Final Engineering Design Report (FEDR)

- Documentation of optimisation methodology (e.g., algorithms used, constraints applied).
- Sensitivity analysis results for key hyperparameters and performance metrics.
- Trade-off studies between payload, range for fleet fuel burn, and DOC.
- Summary of integration strategy and version control system.
- Discussion of robustness of the final design point under uncertainty.

Poster Presentation

- Visual representation of optimisation workflow and key trade-offs.
- Graphs showing Pareto fronts or sensitivity plots for primary metrics (fuel burn, DOC, ATP).
- Highlight cross-disciplinary interactions and how they influenced the final design.
- Summary of integration challenges and solutions.

2.7.3 Getting Started

Begin by reviewing the structure of the supplied MDO tool and understanding how inputs and outputs interact. Before diving into complex algorithms, explore basic optimisation strategies such as Monte Carlo sampling for design space exploration and gradient-based methods for local improvements. These approaches will help you understand sensitivities and trade-offs early on. Once familiar, try these techniques in the MDO tool using simple objectives, such as block fuel, then progressively incorporate additional metrics, such as DOC and climate impact.

2.8 Landing Gear Engineer

The landing gear engineer is responsible for the overall sizing and placement of the main landing gear system and nose landing gear system.

2.8.1 Responsibilities

Concept Down Selection:

- Extract critical mission parameters from requirements.
- Study a unique configuration and rank according to chosen parameters.
- Engage with specialism material and explore the supplied MDO tool.

MDO Development: For the chosen hyperparameters and configurations:

- Develop a class I or II methodology to estimate the mass of the main landing gear and nose landing gear systems:
 - Define the number of wheels required for the nose landing gear and main landing gear. Done in collaboration with *Mass & Stability Engineer* (MTOW estimate)
 - programmatically estimating the required longitudinal and lateral placement of the landing gear respecting ground stability requirements as provided in the specialist material. Done in collaboration with the *Mass & Stability Engineer* (COM range).
 - Estimate the required length of the landing gear (compressed) to achieve required engine clearance and tail strike angle as provided in the specialist material. Done in collaboration with *PPC Engineer* (Engine Placement) and *Mass & Stability Engineer* (COM range).

MDO Refinement:

- Refine the Class II.5 model (physics-based model) for landing gear mass estimation.
 - Estimate the required stroke length for the main landing gear as a function of the landing reaction factor λ .
 - Collaborate with the *Wing Structures Engineer* to ensure the critical landing reaction factor is accounted for in the 'landing load case'.
 - Estimate the main landing gear mass based on the the resulting stroke length, shock absorber piston diameter and suitable material choice.

2.8.2 Deliverables

Final Engineering Design Report (FEDR)

- Documentation of class I/II methods used to compare the two configurations
- Documentation of class II.5 method used estimate final configuration.
- Provide sensitivity studies around the final design point.

- Provide shock absorber sizing, main landing gear mass and configuration of nose landing gear and main landing gear at the final design point.

Poster Presentation

- Methodology of The class I/II & II.5 methods used to estimate landing gear mass of the final configuration.
- Comparison of the I/II and II.5 estimations across a range of hyperparameters.
- Student Selected Cross-discipline sensitivity/trade study.

2.8.3 Getting Started

The specialist material will provide additional landing gear system requirements regarding lateral stability of the aircraft on the ground ("turnover angle") and requirements on the load share between the nose landing gear and the main landing gear to support longitudinal positioning of the landing gear. Similarly for the MDO development phase, focus on the geometrical positioning of the landing gear to satisfy geometrical requirements for engine and tail clearance. During MDO refinement you will be introduced to physics-based models to size the required length of the landing gear.

Selecting the number of main landing gear wheels, their arrangement, and tyre size will be challenging during the MDO development phase due to a concept known as "flotation". This is essentially how the aircraft mass is distributed across the tyres of the main landing gear and requires consideration of tyre inflation pressure, contact patch and airport pavement flexibility. This is computed using a dedicated Federal Aviation Administration tool known as COMFAA. Consequently, to get started for the class I or II methodology it is recommended you consider the typical aircraft mass per tyre that is observed for existing aircraft to guide your selection of the number of wheels on your main landing gear. Please note that civil aircraft must have two wheels per axle.

References

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